

2.9 Curves

We can use two approaches to draw curved lines. One approach is to use a curve generation algorithm such as DDA. In this approach a true curve is created. In the second approach the curve is approximated by a number of small straight line segments. This can be achieved with the help of interpolation techniques.

2.9.1 Circular Arc Generation using DDA Algorithm

Digital differential analyzer algorithm uses the differential equation of the curve. The differential equations for simple curve such as circle is fairly easy to solve and we have already discussed it in the chapter 1. Let us see the DDA algorithm for generating circular arcs. The equation for an arc in the angle parameters can be given as,

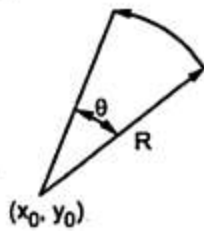


Fig. 2.34

$$\begin{aligned}x &= R \cos \theta + x_0 \\y &= R \sin \theta + y_0\end{aligned}\quad \dots (1)$$

where (x_0, y_0) is the centre of curvature, and R is the radius of arc. (See Fig. 2.34).

Differentiating equation 1 we get,

$$\begin{aligned}dx &= -R \sin \theta d\theta \\dy &= R \cos \theta d\theta\end{aligned}\quad \dots (2)$$

From equation (1) we can solve for $R \cos \theta$ and $R \sin \theta$ as follows.

$$\begin{aligned}x &= R \cos \theta + x_0 \\ \therefore R \cos \theta &= x - x_0 \text{ and} \\ R \sin \theta &= y - y_0\end{aligned}\quad \dots (3)$$

Substituting values of $R \cos \theta$ and $R \sin \theta$ from equation (3) in equation (2) we get,

$$\begin{aligned}dx &= -(y - y_0) d\theta \text{ and} \\ dy &= (x - x_0) d\theta\end{aligned}\quad \dots (4)$$

The values of dx and dy indicate the increment in x and y increment, respectively, to be added in the current point on the arc to get the next point on the arc. Therefore, we can write,

$$\begin{aligned}x_2 &= x_1 + dx = x_1 - (y_1 - y_0) d\theta \\ y_2 &= y_1 + dy = y_1 + (x_2 - x_0) d\theta\end{aligned}\quad \dots (5)$$

The equation (5) forms the basis for arc generation algorithm. From equation 5 we can see that the next point on the arc is the function of $d\theta$. To have a smooth curve, the neighbouring points on the arc should be close to each other. To achieve this, the value of

$d\theta$ should be small enough not to leave gaps in the arc. Usually, the value of $d\theta$ can be determined from the following equation.

$$d\theta = \text{Min} (0.01, 1/(3.2 \times (|x - x_0| + |y - y_0|)))$$

Algorithm

1. Read the centre of a curvature, say (x_0, y_0) .
2. Read the arc angle, say θ .
3. Read the starting point of the arc, say (x, y) .
4. Calculate $d\theta$.

$$d\theta = \text{Min} (0.01, 1 / (3.2 \times (|x - x_0| + |y - y_0|)))$$
5. Initialize Angle = 0.
6. While (Angle < θ)
do
{ Plot (x, y)
 $x = x - (y - y_0) \times d\theta$
 $y = y + (x - x_0) \times d\theta$
 Angle = Angle + $d\theta$
}
7. Stop.

Problems in True-Curve Generation Approach

1. To specify a curve, we need more information than just its end points.
2. It is difficult to apply transformations. For example, a circle when scaled in only one direction becomes an ellipse. If our algorithm supports only circular arc generation then ability to scale pictures is limited.
3. New clipping algorithm is required to clip arcs.
4. The curve generation algorithms for curves other than circular or elliptical such as airplane wings or cars or human faces, are complex.

2.10 Interpolation

In the last section we have seen limitations of true curve generation approach. Furthermore in practice we have to deal with some complex curves for which no direct mathematical function is available. Such curves can be drawn using approximation methods. If we have set of sample points which lie on the required curve, then we can draw the required curve by filling portions of the curve with the pieces of known curves which pass through nearby sample points. The gap between the sample points can be

filled by finding the co-ordinates of the points along the known approximating curve and connecting these points with line segments as shown in the Fig. 2.35.

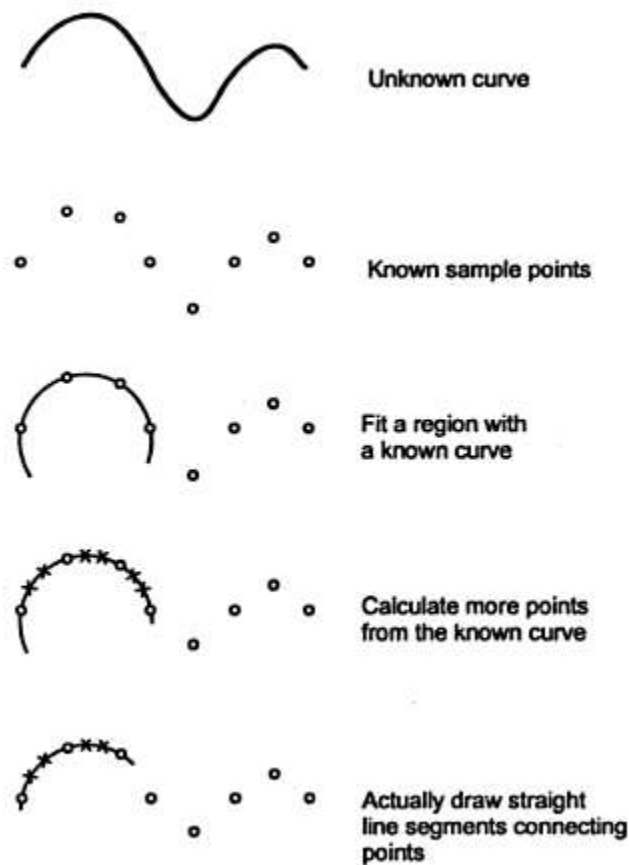


Fig. 2.35 The interpolation process

The main task in this process is to find the suitable mathematical expression for the known curve. There are polynomial, trigonometric, exponential and other classes of functions that can be used to approximate the curve. Usually polynomial functions in the parametric form are preferred. The polynomial functions in the parametric form can be given as

$$\begin{aligned} x &= f_x(u) \\ y &= f_y(u) \\ z &= f_z(u) \end{aligned}$$

We can realize from above equations that the difference between 2 and 3 dimensions is just the addition of the third equation for z . Furthermore the parametric form treats all the three dimensions equally and allows multiple values (several values of y or z for a given x value). Due to these multiple values curves can double back or even cross themselves, as shown in Fig. 2.36.

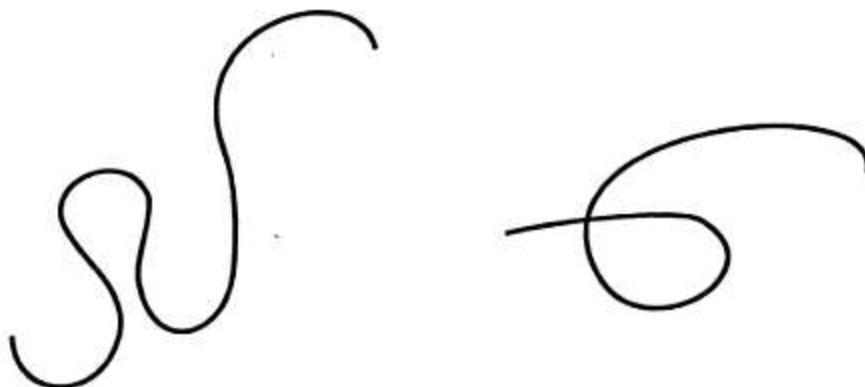


Fig. 2.36 Representation of curves with double back or crossing themselves

We have seen that, we have to draw the curve by determining the intermediate points between the known sample points. This can be achieved using interpolation techniques. Let us see the interpolation process.

Suppose we want a polynomial curve that will pass through n sample points.

$$(x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_n, y_n, z_n)$$

We will construct the function as the sum of terms, one term for each sample point. These functions can be given as,

$$f_x(u) = \sum_{i=1}^n x_i B_i(u)$$

$$f_y(u) = \sum_{i=1}^n y_i B_i(u)$$

$$f_z(u) = \sum_{i=1}^n z_i B_i(u)$$

The function $B_i(u)$ is called 'blending function'. For each value of parameter u , the blending function determines how much the i^{th} sample point affects the position of the curve. In other words we can say that each sample point tries to pull the curve in its direction and the function $B_i(u)$ gives the strength of the pull. If for some value of u , $B_i(u) = 1$ for unique value of i (i.e. $B_i(u) = 0$ for other values of i) then i^{th} sample point has complete control of the curve and the curve will pass through i^{th} sample point. For different value of u , some other sample point may have complete control of the curve. In such case the curve will pass through that point as well. In general, the blending functions give control of the curve to each of the sample points in turn for different values of u . Let us assume that the first sample point (x_1, y_1, z_1) has complete control when $u = -1$, the second when $u = 0$, the third when $u = 1$, and so on. i.e.

When $u = -1 \Rightarrow B_1(u) = 1$ and 0 for $u = 0, 1, 2, \dots, n-2$

When $u = 0 \Rightarrow B_2(u) = 1$ and 0 for $u = -1, 1, \dots, n-2$

:
:
:

When $u = (n-2) \Rightarrow B_n(u) = 1$ and 0 for $u = -1, 0, \dots, (n-1)$

To get $B_1(u) = 1$ at $u = -1$ and 0 for $u = 0, 1, 2, \dots, n-2$, the expression for $B_1(u)$ can be given as,

$$B_1(u) = \frac{u(u-1)(u-2) \dots [u-(n-2)]}{(-1)(-2) \dots (1-n)}$$

where denominator term is a constant used. In general form i^{th} blending function which is 1 at $u = i-2$ and 0 for other integers can be given as :

$$B_i(u) = \frac{(u+1)(u)(u-1) \dots [u-(i-3)][u-(i-1)] \dots [u-(i-2)]}{(i-1)(i-2)(i-3) \dots (1)(-1) \dots (i-n)}$$

The approximation of the curve using above expression is called **Lagrange interpolation**. From the above expression blending functions for four sample points can be given as,

$$B_1(u) = \frac{u(u-1)(u-2)}{(-1)(-2)(-3)}$$

$$B_2(u) = \frac{(u+1)(u-1)(u-2)}{1(-1)(-2)}$$

$$B_3(u) = \frac{(u+1)u(u-2)}{(2)(1)(-1)}$$

$$B_4(u) = \frac{(u+1)u(u-1)}{(3)(2)(1)}$$

Using above blending functions, the expression for the curve passing through sampling points can be realized as follows :

$$x = x_1 B_1(u) + x_2 B_2(u) + x_3 B_3(u) + x_4 B_4(u)$$

$$y = y_1 B_1(u) + y_2 B_2(u) + y_3 B_3(u) + y_4 B_4(u)$$

$$z = z_1 B_1(u) + z_2 B_2(u) + z_3 B_3(u) + z_4 B_4(u)$$

It is possible to get intermediate points between two sampling points by taking values of u between the values of u related to the two sample points under consideration. For example, we can find the intermediate points between second and third sample points for which values of u are 0 and 1, respectively; by taking values of u between 0 and 1. This is shown in Fig. 2.37.

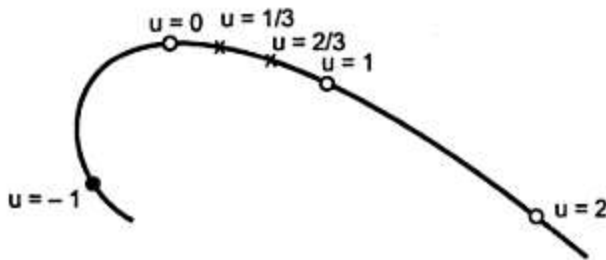


Fig. 2.37 Determining intermediate points for approximation of curve

The subsequent intermediate points can be obtained by repeating the same procedure. Finally the points obtained by this procedure are joined by small straight line segments to get the approximated curve.

Initially, sample points (1, 2, 3, 4) are considered and intermediate points between (2, 3) are obtained. Then

sample point at one end is discarded and sample point at the other end is added to get new sample points (2, 3, 4, 5). Now the curve between sample points (3, 4) is approximated. The subsequent intermediate points can be obtained by repeating the same procedure. The initial and final portions of the curve require special treatment. For the first four points (1, 2, 3, 4) we have to draw region between points (1, 2) with u values between -1 and 0 .

Similarly the blending function for very last step of the curve should be evaluated with u values between 1 and 2 .

Interpolating Algorithm

1. Get the sample points.
2. Get intermediate values of u to determine intermediate points.
3. Calculate blending function values for middle section of the curve.
4. Calculate blending function values for first section of the curve.
5. Calculate blending function values for the last section of the curve.
6. Multiply the sample points by blending functions to give points on approximation curve.
7. Connect the neighbouring points using straight line segments.
8. Stop.

2.11 Spline Representation

To produce a smooth curve through a designated set of points, a flexible strip called **spline** is used. Such a spline curve can be mathematically described with a piecewise cubic polynomial function whose first and second derivatives are continuous across the various curve sections. We can specify a spline curve by giving a set of co-ordinate positions, called **control points**, which indicates the general shape of the curve. When polynomial sections are fitted so that the curve passes through all control points, as shown in the Fig. 2.38 (a), the resulting curve is said to **interpolate** the set of control points. On the other hand, when the polynomials are fitted to the path which is not necessarily passing through all control points, the resulting curve is said to **approximate** the set of control points. This is illustrated in the Fig. 2.38 (b).



Fig. 2.38

2.11.2 Spline Specifications

There are three basic ways of specifying spline curve :

- We can state the set of boundary conditions that are imposed on the spline.
- We can state the matrix that characteristics the spline or
- We can state the set of blending functions that calculate the positions along the curve path by specifying combination of geometric constraints on the curve.

Why to use cubic polynomials ?

Polylines and polygons are first-degree, piecewise linear approximation to curves and surfaces, respectively. But this lower degree polynomials give too little flexibility in controlling the shape of the curve. The higher-degree polynomials give reasonable design flexibility, but introduce unwanted wiggles and also require more computation. For this reason the third-degree polynomials are most often used for representation of curves. These polynomials are commonly known as cubic polynomials.

We can describe the parametric cubic polynomial that is to be fitted between each pair of control points with the following set of equations :

$$\begin{aligned}x(u) &= ax u^3 + bx u^2 + cx u + dx \\y(u) &= ay u^3 + by u^2 + cy u + dy \\z(u) &= az u^3 + bz u^2 + cz u + dz \quad (0 \leq u \leq 1) \quad \dots (1)\end{aligned}$$

For each of these three equations, we need to determine the values of the four coefficients a, b, c and d in the polynomial representation for each of the n curve sections between the n + 1 control points. We do this by setting enough boundary conditions at the joints between curve sections so that we can obtain numerical values for all the coefficient. Let us see the common methods for setting the boundary conditions for cubic interpolation splines.

2.12 Bezier Curves

Bezier curve is another approach for the construction of the curve. A Bezier curve is determined by a defining polygon. Bezier curves have a number of properties that make them highly useful and convenient for curve and surface design. They are also easy to implement. Therefore Bezier curves are widely available in various CAD systems and in general graphic packages. In this section we will discuss the cubic Bezier curve. The reason for choosing cubic Bezier curve is that they provide reasonable design flexibility and also avoid the large number of calculations.

Properties of Bezier curve

1. The basis functions are real.
2. Bezier curve always passes through the first and last control points i.e. curve has same end points as the guiding polygon.
3. The degree of the polynomial defining the curve segment is one less than the number of defining polygon point. Therefore, for 4 control points, the degree of the polynomial is three, i.e. cubic polynomial.
4. The curve generally follows the shape of the defining polygon.
5. The direction of the tangent vector at the end points is the same as that of the vector determined by first and last segments.
6. The curve lies entirely within the convex hull formed by four control points.
7. The convex hull property for a Bezier curve ensures that the polynomial smoothly follows the control points.
8. The curve exhibits the variation diminishing property. This means that the curve does not oscillate about any straight line more often than the defining polygon.
9. The curve is invariant under an affine transformation.

In cubic Bezier curve four control points are used to specify complete curve. Unlike the B-spline curve, we do not add intermediate points and smoothly extend Bezier curve, but we pick four more points and construct a second curve which can be attached to the first.

The second curve can be attached to the first curve smoothly by selecting appropriate control points.

Fig. 2.38 shows the Bezier curve and its four control points. As shown in the Fig. 2.39, Bezier curve begins at the first control point and ends at the fourth control point. This means that if we want to connect two Bezier curves, we have to make the first control point of the second Bezier curve match the last control point of the first curve. We can also observe that at the start of the curve, the curve is tangent to the line connecting first and second control points. Similarly at the end of curve, the curve is tangent to the line connecting the third and fourth control point. This means that, to join two Bezier curves smoothly we have to place the third and the fourth control points of the first curve on the same line specified by the first and the second control points of the second curve.

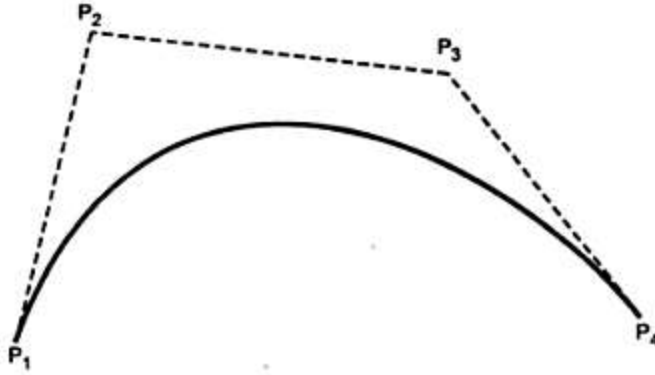


Fig. 2.39 A cubic Bezier spline

The Bezier matrix for periodic cubic polynomial is

$$M_B = \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore P(u) = U \cdot M_B \cdot G_B$$

where

$$G_B = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{bmatrix}$$

and the product $P(u) = U \cdot M_B \cdot G_B$ is

$$P(u) = (1-u)^3 P_1 + 3u(1-u)^2 P_2 + 3u^2(1-u) P_3 + u^3 P_4$$

Therefore, the four blending functions for cubic Bezier curve are :

$$BEZ_{0,3}(u) = (1-u)^3$$

$$BEZ_{1,3}(u) = 3u(1-u)^2$$

$$BEZ_{2,3}(u) = 3u^2(1-u)$$

$$BEZ_{3,3}(u) = u^3$$

► **Example 2.5 :** Construct the Bezier curve of order 3 and with 4 polygon vertices $A(1, 1)$, $B(2, 3)$, $C(4, 3)$ and $D(6, 4)$.

Solution : The equation for the Bezier curve is given as

$$P(u) = (1-u)^3 P_1 + 3u(1-u)^2 P_2 + 3u^2(1-u) P_3 + u^3 P_4$$

... for $0 \leq u \leq 1$

where $P(u)$ is the point on the curve P_1, P_2, P_3, P_4 .

Let us take $u = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$

$$\therefore P(0) = P_1 = (1, 1)$$

$$\begin{aligned}\therefore P\left(\frac{1}{4}\right) &= \left(1 - \frac{1}{4}\right)^3 P_1 + 3\frac{1}{4}\left(1 - \frac{1}{4}\right)^2 P_2 + 3\left(\frac{1}{4}\right)^2\left(1 - \frac{1}{4}\right) P_3 + \left(\frac{1}{4}\right)^3 P_4 \\ &= \frac{27}{64}(1, 1) + \frac{27}{64}(2, 3) + \frac{9}{64}(4, 3) + \frac{1}{64}(6, 4) \\ &= \left[\frac{27}{64} \times 1 + \frac{27}{64} \times 2 + \frac{9}{64} \times 4 + \frac{1}{64} \times 6, \quad \frac{27}{64} \times 1 + \frac{27}{64} \times 3 + \frac{9}{64} \times 3 + \frac{1}{64} \times 4\right] \\ &= \left[\frac{123}{64}, \frac{139}{64}\right] \\ &= (1.9218, 2.1718)\end{aligned}$$

$$\begin{aligned}\therefore P\left(\frac{1}{2}\right) &= \left(1 - \frac{1}{2}\right)^3 P_1 + 3\frac{1}{2}\left(1 - \frac{1}{2}\right)^2 P_2 + 3\left(\frac{1}{2}\right)^2\left(1 - \frac{1}{2}\right) P_3 + \left(\frac{1}{2}\right)^3 P_4 \\ &= \frac{1}{8}(1, 1) + \frac{3}{8}(2, 3) + \frac{3}{8}(4, 3) + \frac{1}{8}(6, 4) \\ &= \left[\frac{1}{8} \times 1 + \frac{3}{8} \times 2 + \frac{3}{8} \times 4 + \frac{1}{8} \times 6, \quad \frac{1}{8} \times 1 + \frac{3}{8} \times 3 + \frac{3}{8} \times 3 + \frac{1}{8} \times 4\right] \\ &= \left[\frac{25}{8}, \frac{23}{8}\right] \\ &= (3.125, 2.875)\end{aligned}$$

$$\begin{aligned}\therefore P\left(\frac{3}{4}\right) &= \left(1 - \frac{3}{4}\right)^3 P_1 + 3\frac{3}{4}\left(1 - \frac{3}{4}\right)^2 P_2 + 3\left(\frac{3}{4}\right)^2\left(1 - \frac{3}{4}\right) P_3 + \left(\frac{3}{4}\right)^3 P_4 \\ &= \frac{1}{64} P_1 + \frac{9}{64} P_2 + \frac{27}{64} P_3 + \frac{27}{64} P_4 \\ &= \frac{1}{64}(1, 1) + \frac{9}{64}(2, 3) + \frac{27}{64}(4, 3) + \frac{27}{64}(6, 4) \\ &= \left[\frac{1}{64} \times 1 + \frac{9}{64} \times 2 + \frac{27}{64} \times 4 + \frac{27}{64} \times 6, \quad \frac{1}{64} \times 1 + \frac{9}{64} \times 3 + \frac{27}{64} \times 3 + \frac{27}{64} \times 4\right] \\ &= \left[\frac{289}{64}, \frac{217}{64}\right] \\ &= (4.5156, 3.375)\end{aligned}$$

$$P(1) = P_4 = (6, 4)$$

The Fig. 2.40 shows the calculated points of the Bezier curve and curve passing through it.

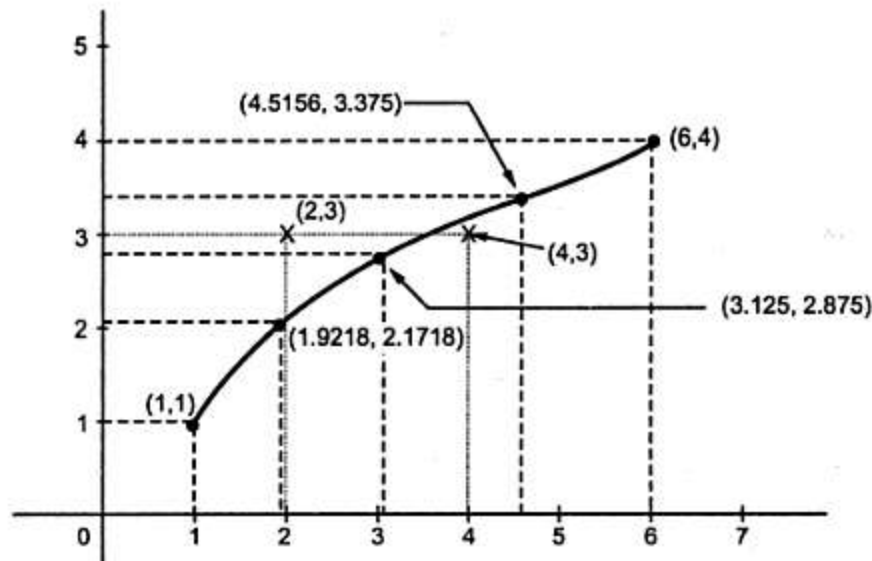


Fig. 2.40 Plotted Bezier curved

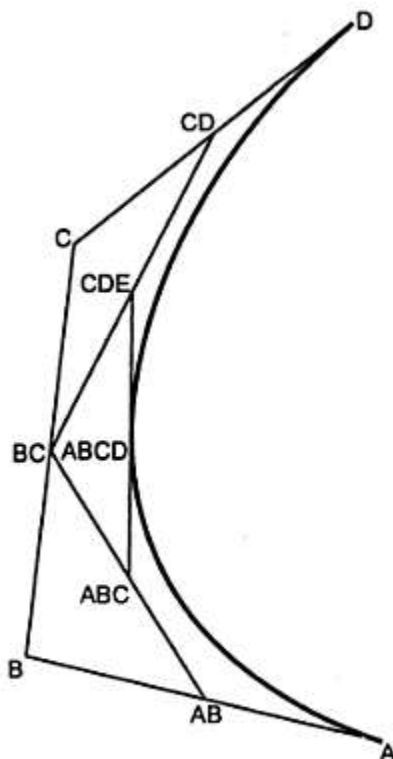


Fig. 2.41 Subdivision of a Bezier spline

Another approach to construct the Bezier curve is called midpoint approach. In this approach the Bezier curve can be constructed simply by taking midpoints. In midpoint approach midpoints of the lines connecting four control points (A, B, C, D) are determined (AB, BC, CD). These midpoints are connected by line segments and their midpoints ABC and BCD are determined. Finally these two midpoints are connected by line segments and its midpoint ABCD is determined. This is illustrated in Fig. 2.41.

The point ABCD on the Bezier curve divides the original curve into two sections. This makes the points A, AB, ABC and ABCD are the control points for the first section and the points ABCD, BCD, CD and D are the control points for the second section. By considering two sections separately we can get two more sections for each separate section i.e. the original Bezier curve gets divided into

four different curves. This process can be repeated to split the curve into smaller sections until we have sections so short that they can be replaced by straight lines or even until the sections are not bigger than individual pixels.

Algorithm

1. Get four control points say A (x_A, y_A), B (x_B, y_B), C (x_C, y_C), D (x_D, y_D).
2. Divide the curve represented by points A, B, C and D in two sections.
$$x_{AB} = (x_A + x_B) / 2$$
$$y_{AB} = (y_A + y_B) / 2$$
$$x_{BC} = (x_B + x_C) / 2$$
$$y_{BC} = (y_B + y_C) / 2$$
$$x_{CD} = (x_C + x_D) / 2$$
$$y_{CD} = (y_C + y_D) / 2$$
$$x_{ABC} = (x_{AB} + x_{BC}) / 2$$
$$y_{ABC} = (y_{AB} + y_{BC}) / 2$$
$$x_{BCD} = (x_{BC} + x_{CD}) / 2$$
$$y_{BCD} = (y_{BC} + y_{CD}) / 2$$
$$x_{ABCD} = (x_{ABC} + x_{BCD}) / 2$$
$$y_{ABCD} = (y_{ABC} + y_{BCD}) / 2$$
3. Repeat the step 2 for section A, AB, ABC and ABCD and section ABCD, BCD, CD and D.
4. Repeat step 3 until we have sections so short that they can be replaced by straight lines.
5. Replace small sections by straight lines.
6. Stop.